MASTER IN ASTROPHYSICS AND SPACE SCIENCE

Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences

Numerical Simulations of AGN Accretion Disks in PLUTO

CAMK PAN Summer Student Program

Fritz Ali Agildere

October 13, 2026

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics

Disk Classifier
- Passive Tracer
- Construction & Validation

Results
- Relaxation
- Steady State
- Density Waves

Discussion
- Limitations
- Outlook

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial & Equatorial Symmetry
  - Replaced Stellar Surface with Absorbing Diode Inner Boundary
  - Replaced Newtonian with Paczyński–Wiita Potential to Reproduce ISCO

$$\Phi = -\frac{GM}{r - r_s}$$

  - Measured Lengths and Times in Gravitational Units with c for Velocities

$$r_g = GM/c^2 \qquad t_g = r_g/c$$

  - Embedded Kluźniak–Kita Disk from $R = 6$ with $\epsilon = 0.1$ in Hydrostatic Corona
  - Reinjected Analytic Disk at Outer Boundary and Blocked Coronal Inflow

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $1 \times 10^8\ M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\ \mathrm{cm} \qquad t_g \approx 493\ \mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\ \mathrm{g\,cm^{-3}}$$

  - Balanced Initial Viscous Heating and Radiative Cooling at $\alpha = 0.5$
  - Included Kramers Absorption as well as Thomson Scattering Opacities
- Added Effective Compton Opacity since PLUTO Treats Scattering as Coherent

$$\kappa_C = \kappa_{\mathrm{es}} \frac{4 k_B T_r^4}{m_e c^2 \left(T_g + T_r\right)\left(T_g^2 + T_r^2\right)}$$

  - Reproduces Compton Exchange Rate and Prevents Gas Heating up to 1×10^{11} K
  - Rate Capped above 1×10^9 K, Decoupled above $0.9\,c$ and Gas Floor at 1×10^4 K

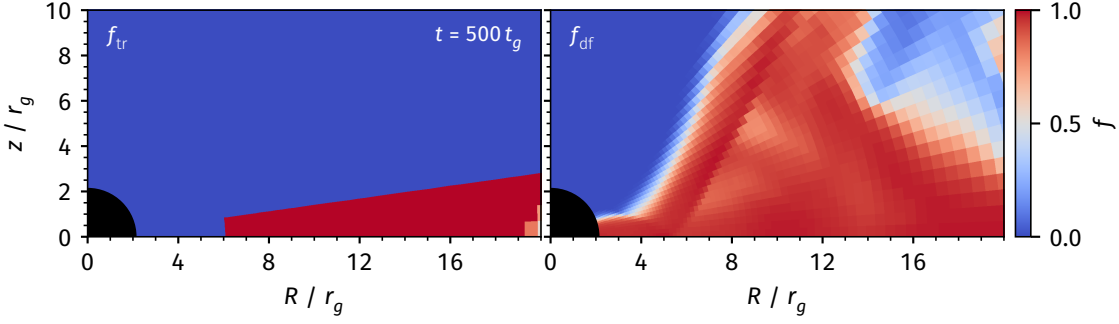
Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling and Opacities to Disk

$$\partial_t q + \mathbf{v} \cdot \nabla q = 0$$

  - Stays at Initial Wedge Once Radiation Enabled while Disk Evolves
  - Cause Not Isolated in Code, Only Restricted Physics Affected
- Replaced Tracer by Classifier Built from Current State of Each Cell Alone
  - Same Disk Condition as Initial Setup, Identical to Tracer at Start
  - Robust to Restarts by Writing Classifier State with Every Output

Radiation Enabled – Tracer & Classifier



Construction

- Classified Cells by `DiskFraction` $f \in [0, 1]$ from Density and Rotation
  - Stored Disk and Corona Reference Densities in 90 Logarithmic Radial Bins
  - Located Cell Density between References on Logarithmic Scale

$$s = \frac{\log \rho - \log \rho_c}{\max\left(\log \rho_d - \log \rho_c, 0.5\right)}$$

  - Multiplied Density and Keplerian Rotation Criteria with Logistic Function

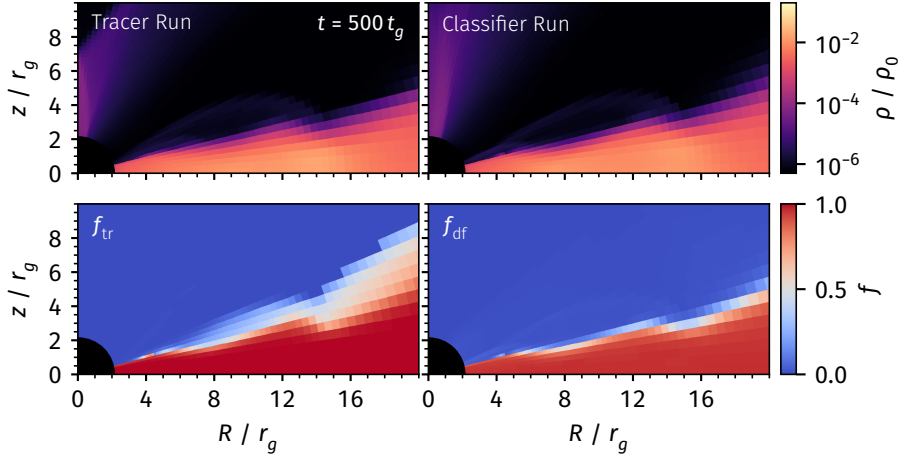
$$f = \sigma\left(\frac{s - 1/6}{0.3}\right)\sigma\left(\frac{|v_\phi|\sqrt{R} - 0.5}{0.1}\right)$$

  - Updated References as f and $1 - f$ Weighted Means via Moving Average with Weight $w = 1 - \exp(-\Delta t/\tau)$ and Time Constant $\tau = 6\, t_g$

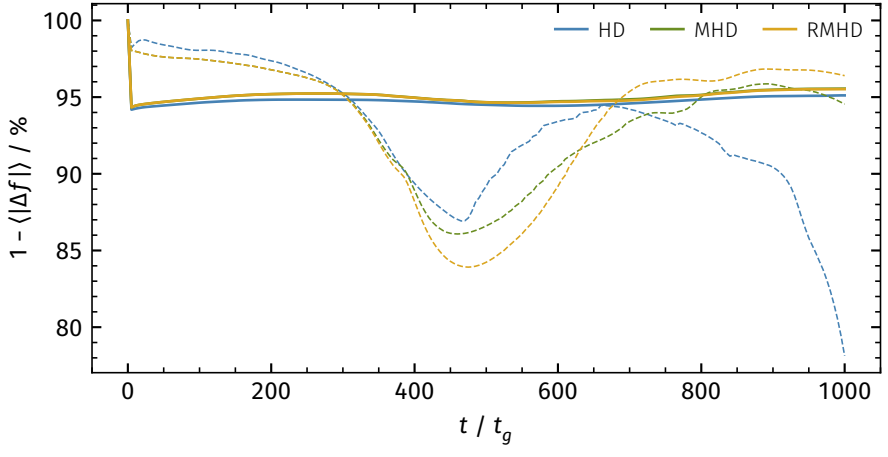
Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification
  - Plunging Gas inside ISCO Rotates Faster than Keplerian and Counted as Disk
- Current Version Repairs Itself instead of Locking Error in Place
  - Enforced Minimum Span of 0.5 dex between Disk and Corona References
  - Copied References from Nearest Valid Bin, Searching Outward First
  - Smoothed with Time Constant for Same Memory at Every Resolution

Validation – Hydrodynamics



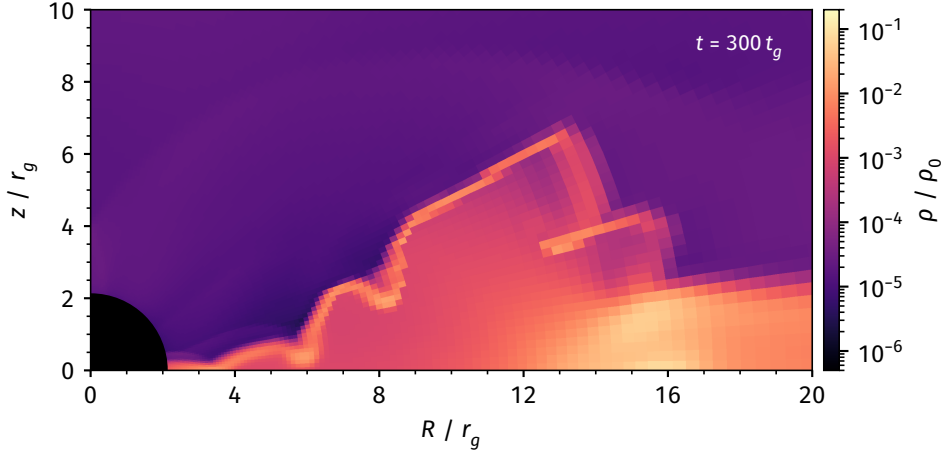
Validation – Agreement



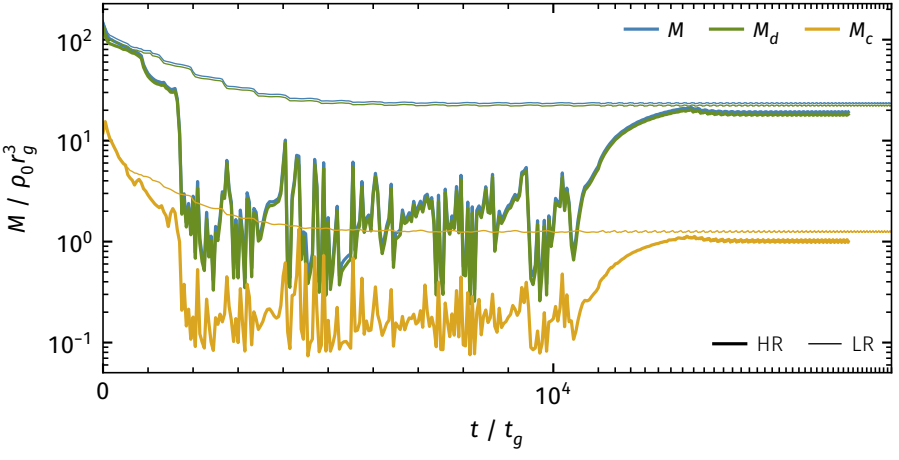
Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every 50 Gravitational Times, Refined to 5 for Last 10 000
- Inflated Disk and Broke up Dense Surface Layer during First Hundreds of t_g
  - HR Disk Blown Apart and Rebuilt by $t \approx 15000$, LR Declined Smoothly
  - Corona Drained Early and Not Replenished due to Blocked Inflow
- Steady State when Disk Mass, Radiation Energy, Accretion Rate and Luminosity Remain within Two Standard Errors of Final Fifth
  - Reached at $t = 23725$ for HR and $t = 9225$ for LR

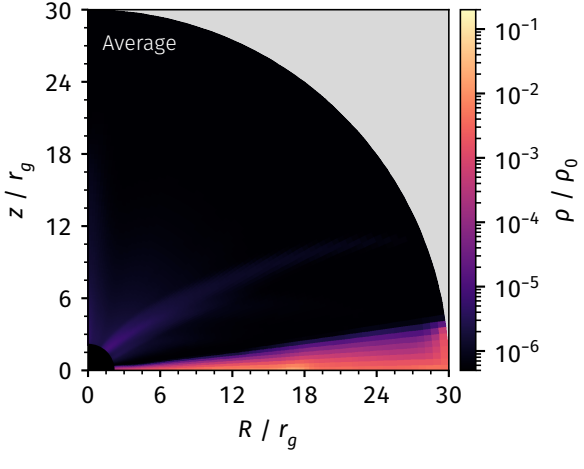
Early Evolution – Density



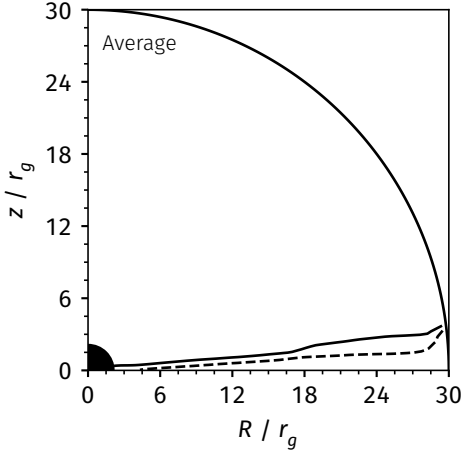
Relaxation – Mass



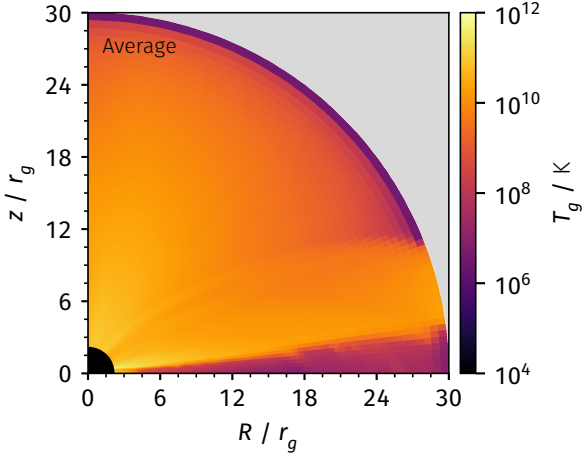
Steady State – Density



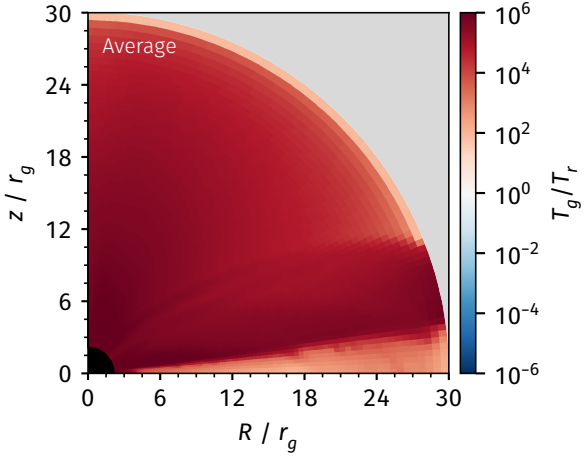
Steady State – Disk Surface



Steady State – Gas Temperature



Steady State – Temperature Ratio



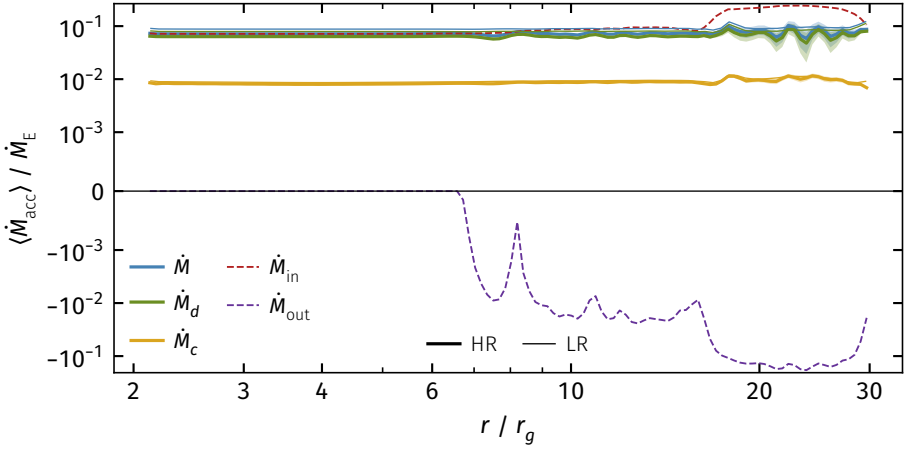
Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Further Out
  - Accretion Rate **0.073** and **0.091** in Eddington Units, about **0.3** $M_\odot\,yr^{-1}$
  - Luminosity **0.056** and **0.070** with Radiative Efficiency $\eta \approx 0.048$
- Surface Density below Steady Thin Disk Profile

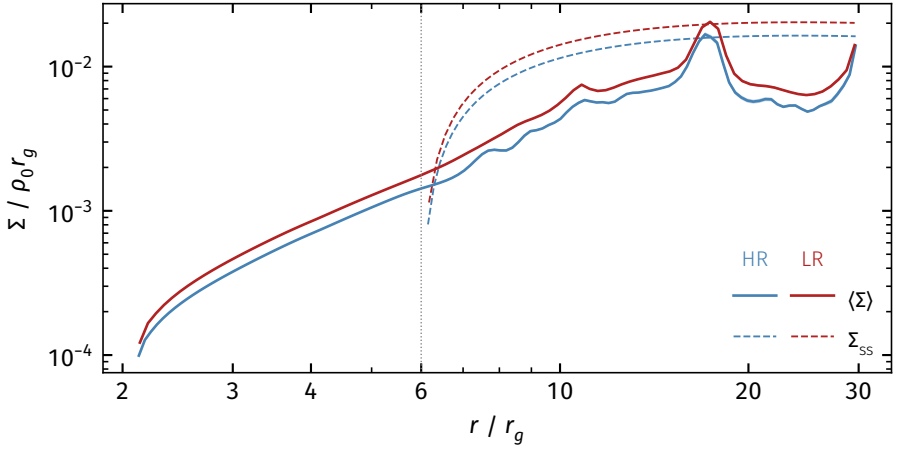
$$\Sigma_{SS} = \frac{\dot{M}}{3\pi\nu}\left(1 - \sqrt{6/R}\right)$$

  - Effective Viscosity Two to Three Times Higher than Prescribed α
- Viscous Heating Balanced by Compton Cooling, Ratio of **0.8** to **0.9**
  - Gas a Hundred to Two Thousand Times Hotter than Radiation at Midplane

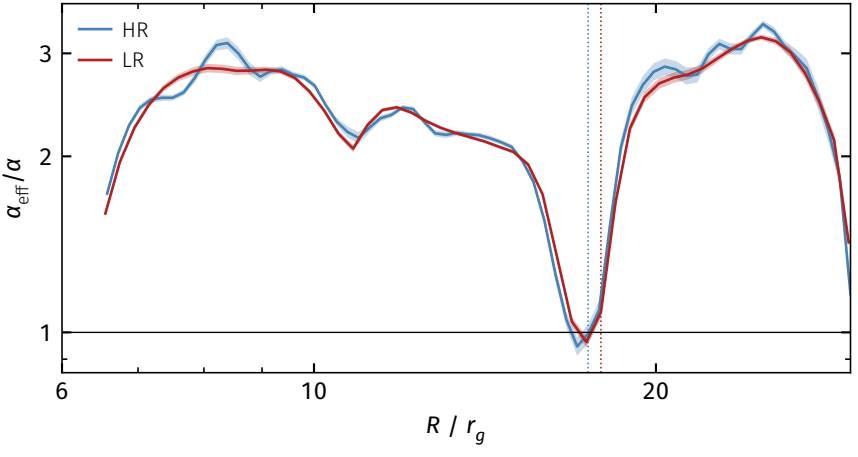
Steady State – Accretion



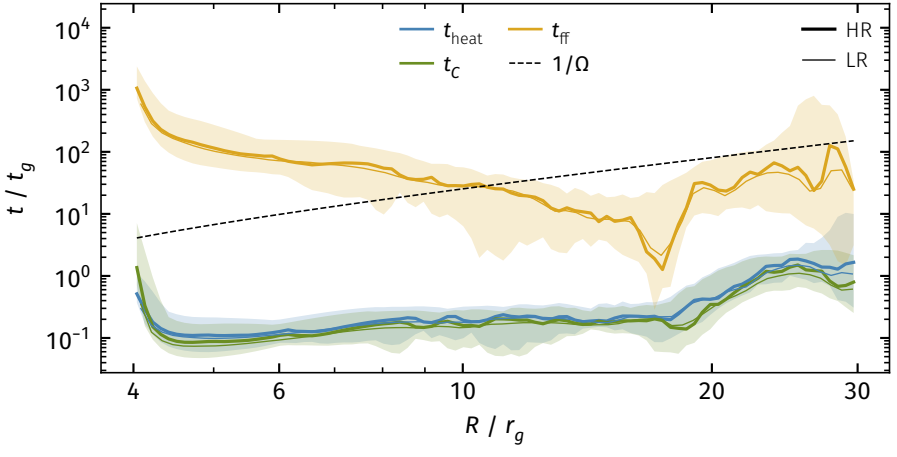
Steady State – Surface Density



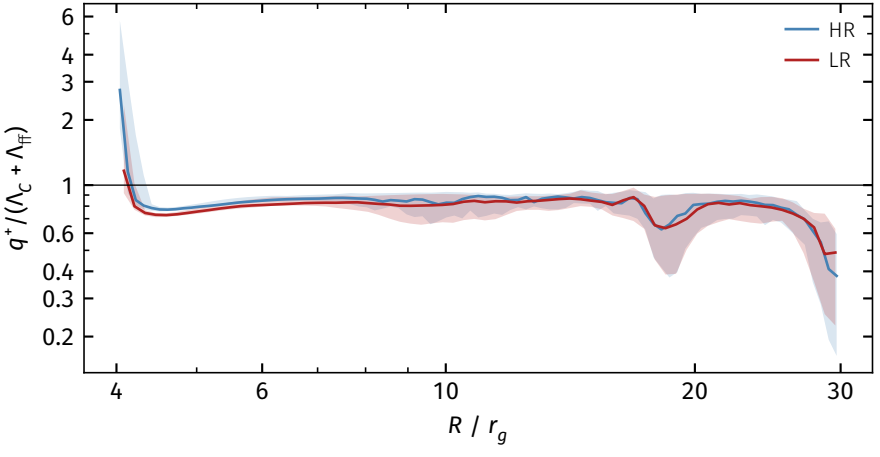
Steady State – Effective Viscosity



Steady State – Heating & Cooling Times



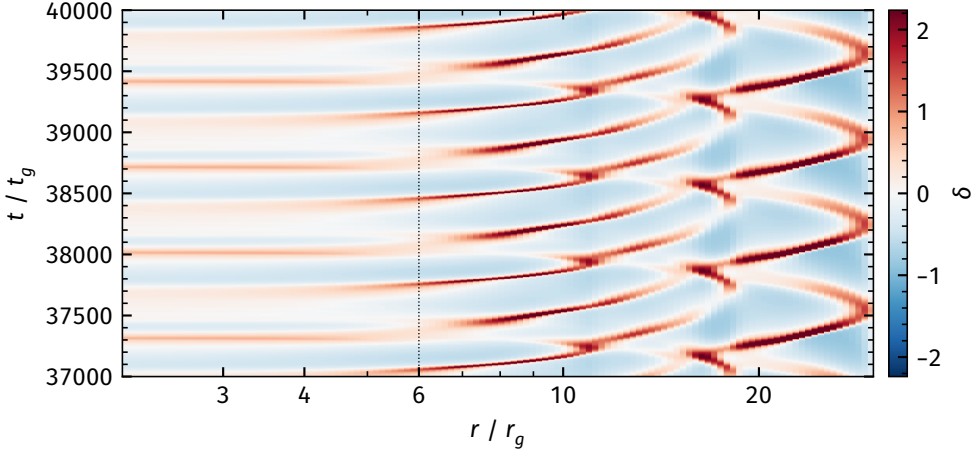
Steady State – Heating & Cooling Ratio



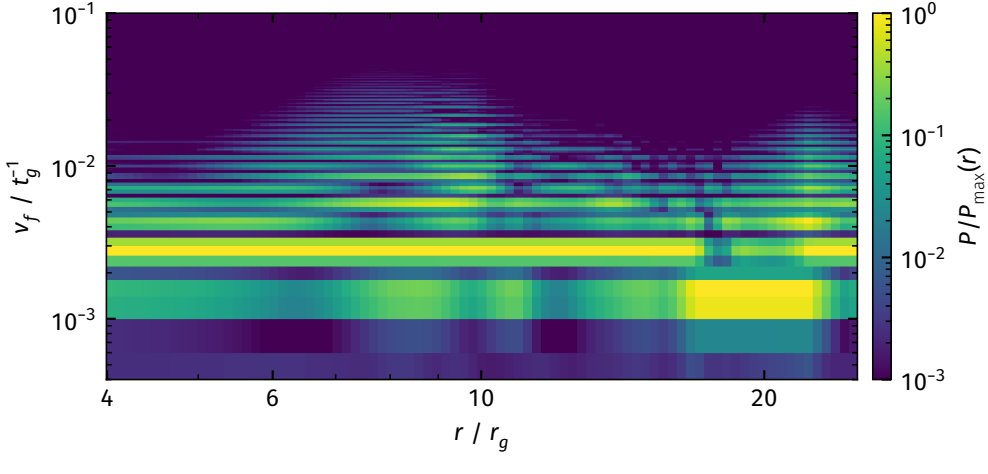
Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**
  - Period of **349** Gravitational Times, about Two Days for Our SMBH
- Phase Speed of $0.019\,c$ Steady between Radii of **4** and **17**
  - **2.2** Times Adiabatic Sound Speed with Radiation, so Not Plain Sound Waves
  - Same Period at Both Resolutions Suggests Origin in Setup rather than Numerics
  - Close to Thermal Time $1/(\alpha\Omega)$ of $330\,t_g$ at Outer Boundary

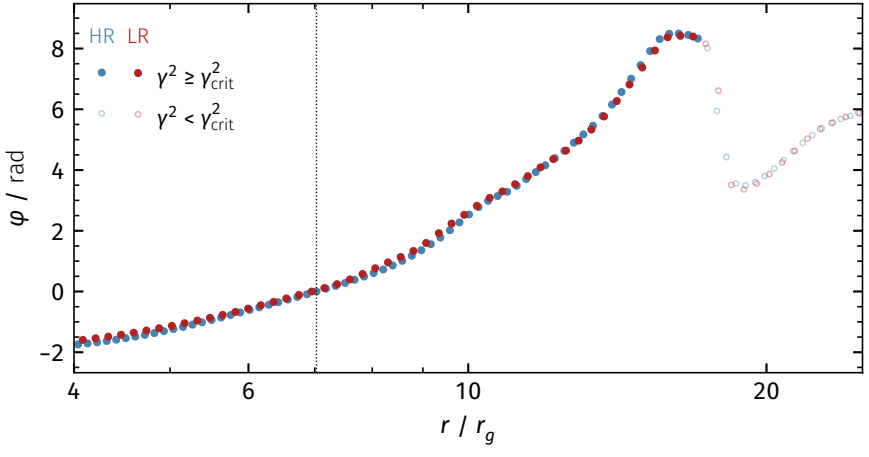
Density Waves – Perturbation



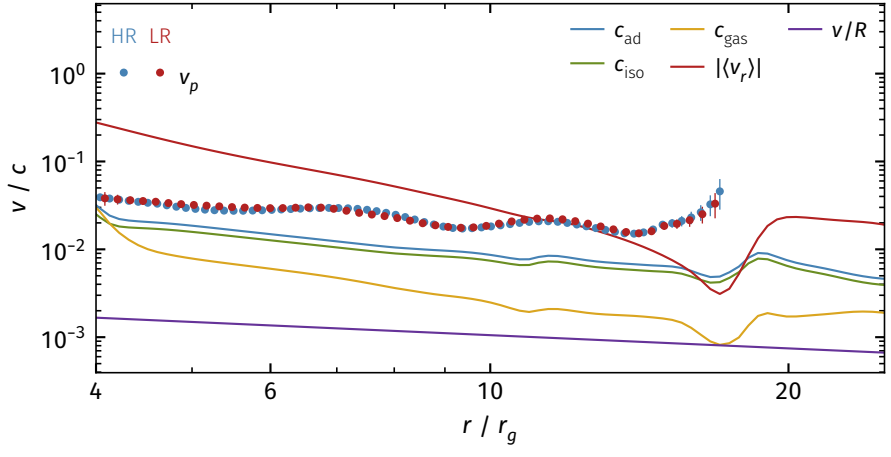
Density Waves – Power Spectrum



Density Waves – Phase



Density Waves – Phase Speed



Resolution Comparison

Quantity	HR	LR
Cells ($r \times \theta$)	120×90	90×45
Classifier Version	Current	First
Steady State Start / t_g	23 725	9225
Accretion Rate / $\dot{M}_E$	0.073	0.091
Luminosity / L_E	0.056	0.070
Efficiency	0.048	0.049
Equilibrium Radius / r_g	17.4	17.9
Aspect Ratio ($R = 10$)	0.057	0.042
Cells per Scale Height	3.3	1.2
Wave Period / t_g	349	349
Phase Speed / c	0.0190	0.0195

Limitations

- Viscosity Uses Sound Speed of Initial Disk, Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability, so Stability No Test
- Boundaries Set Gas Supply and May Cause Inflow Stream and Wave Period
  - Mass Budget Open by 3 % between $r = 2.5$ and 25, 18 % for Full Domain
- Gas Temperatures Likely Too High for Setup to Be Fully Trusted
  - Corona up to 1×10^{11} K with $k_B T$ about 46 Times Electron Rest Energy
  - Neither Relativistic Compton Corrections nor Pair Production Included
- Plunging Gas inside about Four r_g Moves Superluminally in Pseudo Potential
- Axisymmetric Hydrodynamics Replaces MRI Turbulence by α Prescription

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest
  - Start from Standard Thin Disk Solution Suited to Radiation Pressure
- Built Working SMBH Radiation Setup in PLUTO with Effective Compton Coupling
  - `DiskFraction` Agrees with Tracer to **95 %** by Mass and Follows Disk with Radiation
  - Steady States at about a Tenth of Eddington Rate with Compton Balanced Heating
  - Nonlinear Density Waves with Period of about **350** t_g Traveling Outward

Thank You!

- A. Mignone et al., *PLUTO: A Numerical Code for Computational Astrophysics* (ApJS) 2007
- M. Čemeljić, *Atlas of Numerical Solutions for Star-Disk Magnetospheric Interaction* (A&A) 2019
- J. D. Melon Fuksman et al., *A Two-Moment Radiation Hydrodynamics Scheme Applicable to Simulations of Planet Formation in Circumstellar Disks* (ApJ) 2021
- B. Paczyński, P. J. Wiita, *Thick Accretion Disks and Supercritical Luminosities* (A&A) 1980
- Y.-F. Jiang et al., *Super-Eddington Accretion Disks around Supermassive Black Holes* (ApJ) 2019

Resources on GitHub: here

Model Equations

- Initial Disk Pressure and Density from Analytic Solution of Kluźniak & Kita

$$C = \frac{2}{5\epsilon^2}\left(\frac{1}{r} - \frac{1 - 5\epsilon^2/2}{R}\right) \qquad p = \epsilon^2 C^{5/2} \qquad \rho = C^{3/2}$$

- Corona in Hydrostatic Equilibrium with $\rho_c = 0.01$ in Code Units

$$\rho = \rho_c r^{-3/2} \qquad p = 0.4\rho_c r^{-5/2}$$

- Shear Viscosity with Sound Speed of Initial Disk, Restricted by Classifier

$$\eta_v = \frac{2}{3}\alpha\rho\epsilon^2 C R^{3/2} f \qquad v = \frac{2}{3}\alpha\epsilon^2 R^{1/2}$$

- Density Floor at 5×10^{-7} in Code Units with Constant Sound Speed and Momentum

Opacity Model

- Kramers Free Free and Bound Free Absorption with Guillotine Factor

$$\kappa_K = \left(K_{\mathrm{ff}} + K_{\mathrm{bf}}\right) \rho\, T_g^{-7/2} \qquad \kappa_{\mathrm{es}} = 0.2\,(1 + X)$$

- Compton Exchange Held at Cap above $T_{\max} = 1 \times 10^9$ K

$$\kappa_C = \kappa_{\mathrm{es}} \frac{4 k_B T_r^4 \left(T_{\max} - T_r\right)}{m_e c^2 \left(T_g^4 - T_r^4\right)}$$

- Gated by Classifier and Decoupled near Speed of Light, Thomson Scattering on All Gas

$$f_d = \max\left(\frac{f - 0.05}{0.95}, 0\right) \qquad g_v = \left[1 + \exp\left(\frac{\beta - 0.9}{0.03}\right)\right]^{-1}$$

$$\kappa_{\mathrm{abs}} = g_v f_d \left(\kappa_K + \kappa_C\right) \qquad \kappa_{\mathrm{sc}} = g_v \left(\kappa_{\mathrm{es}} - f_d \kappa_C\right)$$

Classifier Details

- Corona Reference as $1 - f$ and Disk Reference as f Weighted Mean Density per Bin
  - Disk Estimate Counts Only above One Thousandth of Bin Volume
  - Bins without Valid Estimate Copy Nearest Valid Bin
- Guards Added after Inversion in First Version
  - Span below 0.5 dex Copies Both References from Nearest Sufficient Bin
  - Disk Reference Raised to Minimum Span if No Such Bin Exists
  - Fixed Weight 0.05 per Step Replaced by Time Constant τ
- Optional Geometric Mode Independent of Classification
  - Disk Reference from Maximum Density, Corona from Geometric Mean above $z/R = 0.5$

Analysis Methods

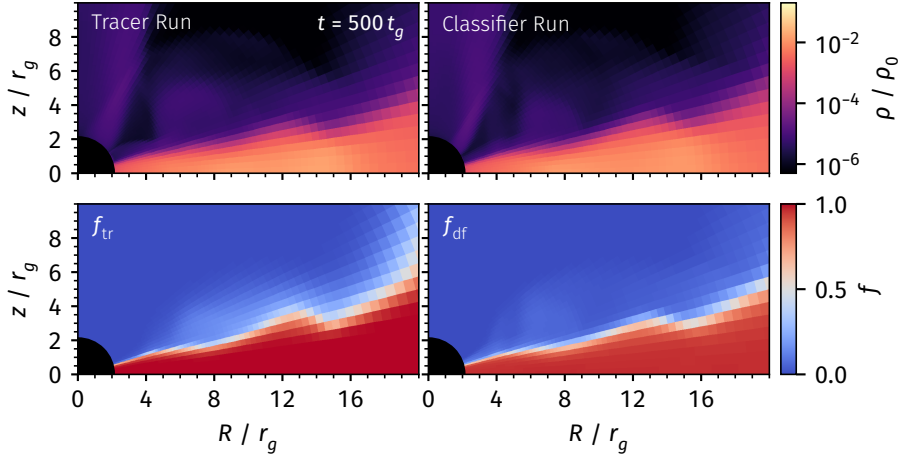
- Standard Errors from Block Averages over **1000** Gravitational Times
- Relaxation from Running Means over $W = 2500$ Compared with Final Fifth

$$D(t_c) = \frac{|\bar{X}_W(t_c) - X_{\mathrm{ref}}|}{\sigma}$$

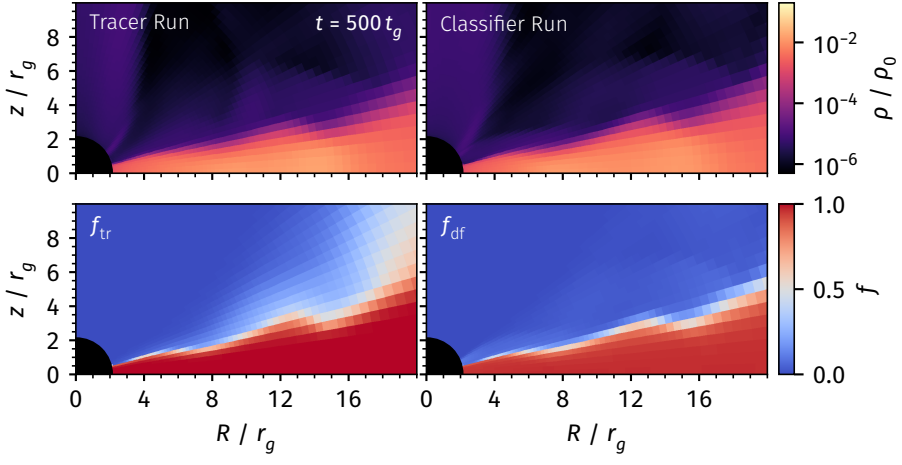
- Inflow Equilibrium where Accretion Rate within **20 %** of Inner Value
- Welch Spectra of Surface Density Perturbation $\delta = \Sigma/\langle\Sigma\rangle - 1$
  - Phase at Peak Frequency Relative to $r = 7$ for Significant Coherence Only
  - Local Phase Speed from Linear Fits over **15 %** in Radius

$$v_p = \omega \,/\, \frac{\mathrm{d}\varphi}{\mathrm{d}r}$$

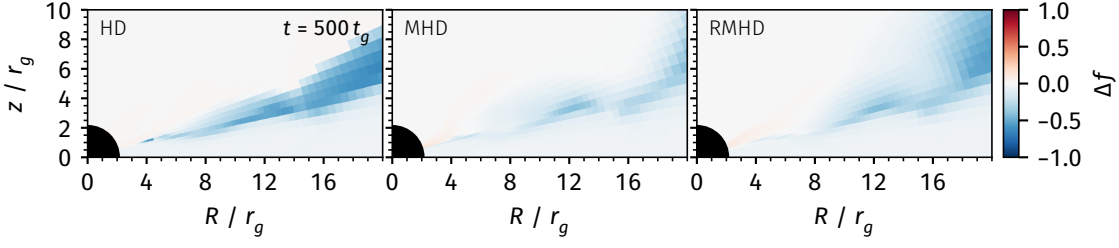
Validation – Magnetohydrodynamics



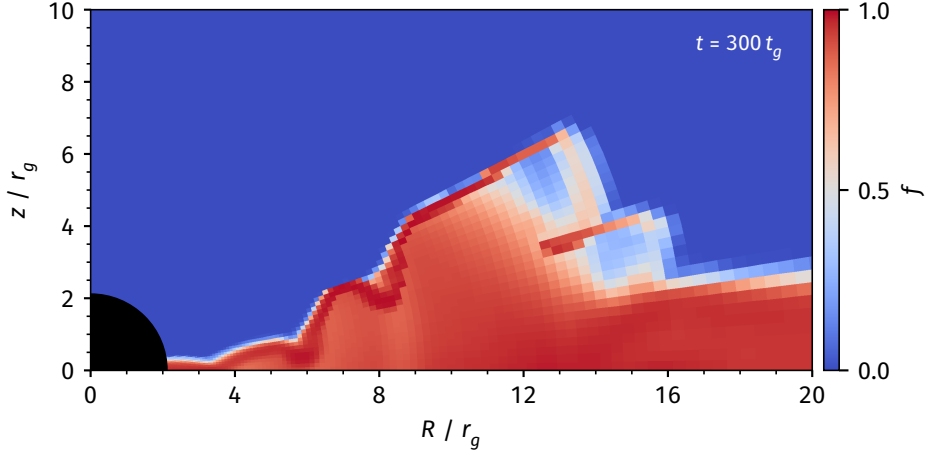
Validation – Resistive Magnetohydrodynamics



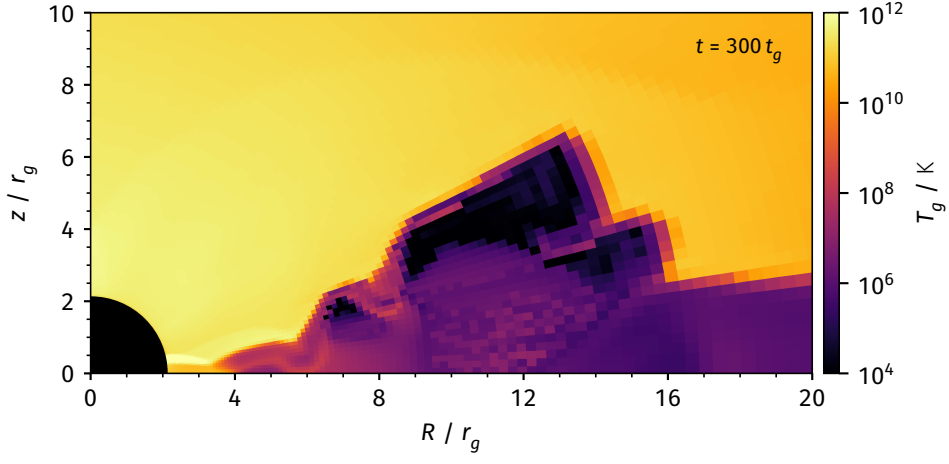
Validation – Classifier Minus Tracer



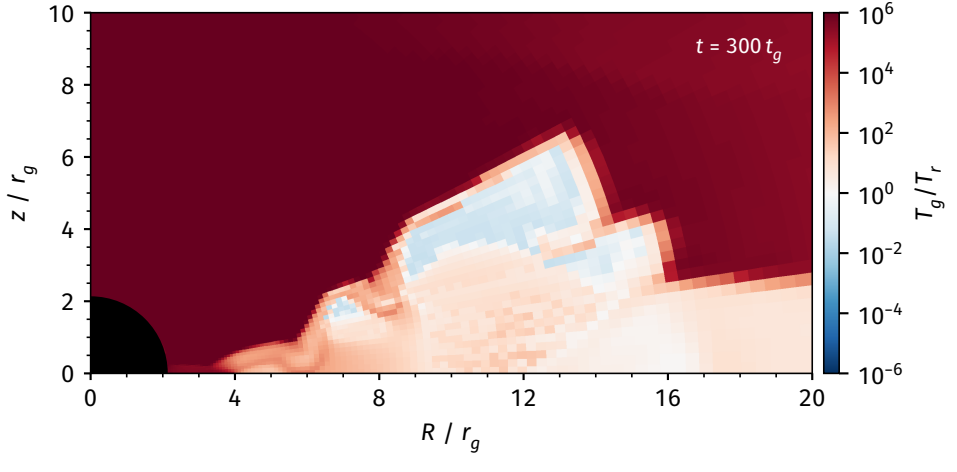
Early Evolution – DiskFraction



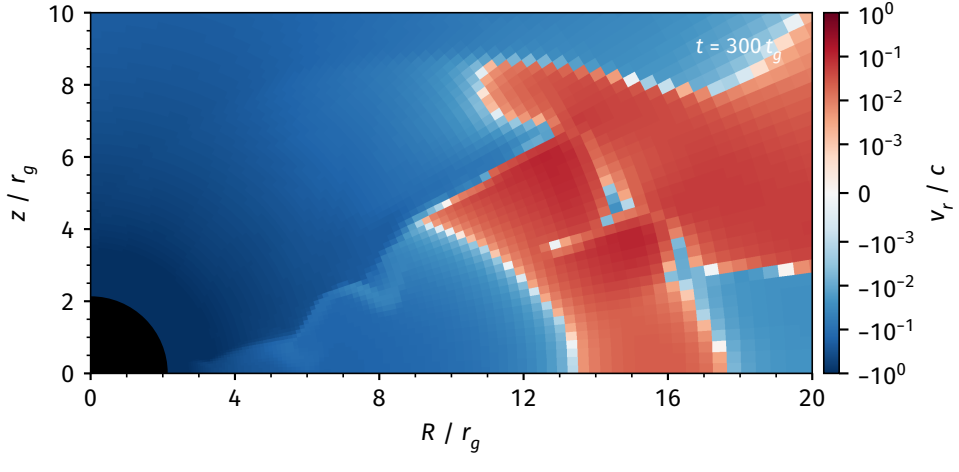
Early Evolution – Gas Temperature



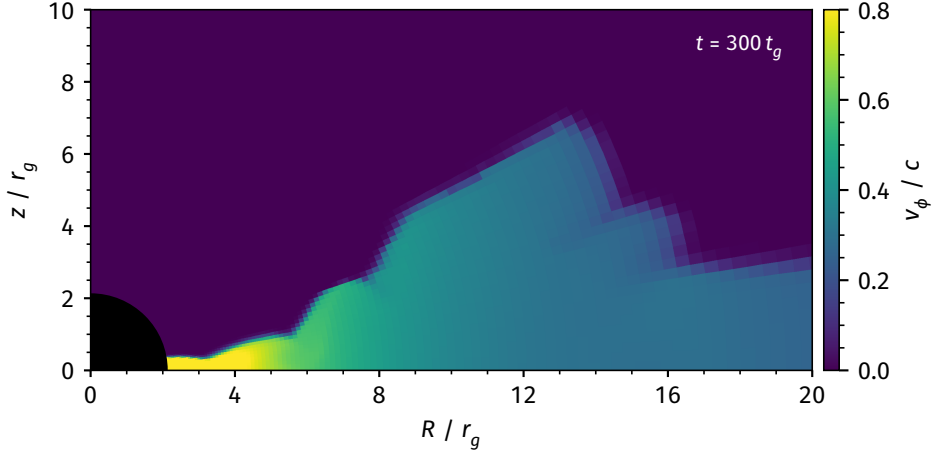
Early Evolution – Temperature Ratio



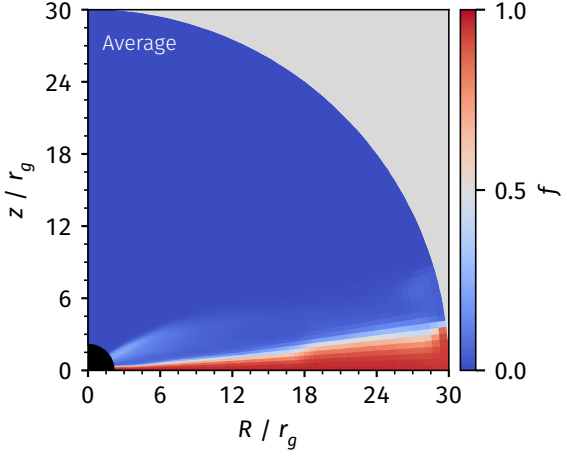
Early Evolution – Radial Velocity



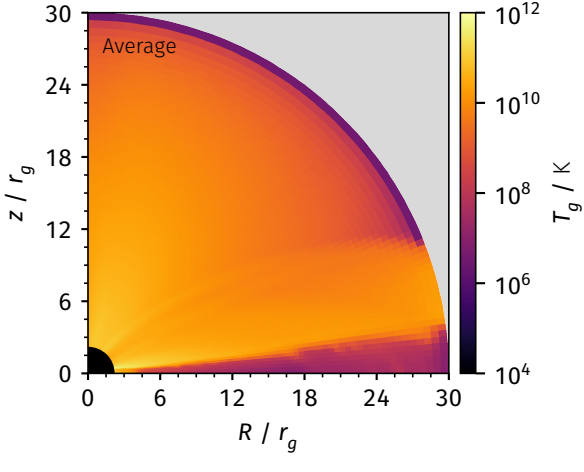
Early Evolution – Rotation Velocity



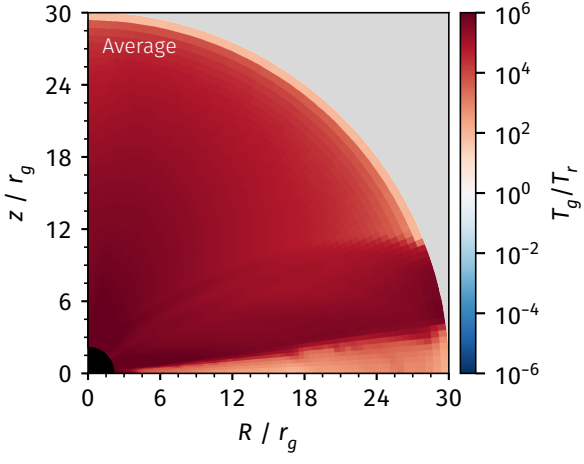
Steady State – DiskFraction



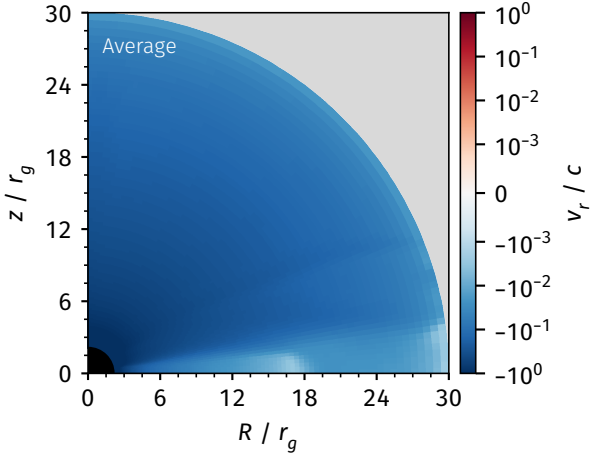
Steady State – Gas Temperature



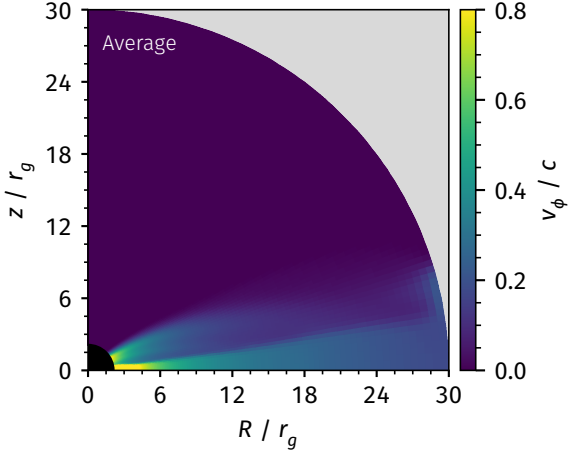
Steady State – Temperature Ratio



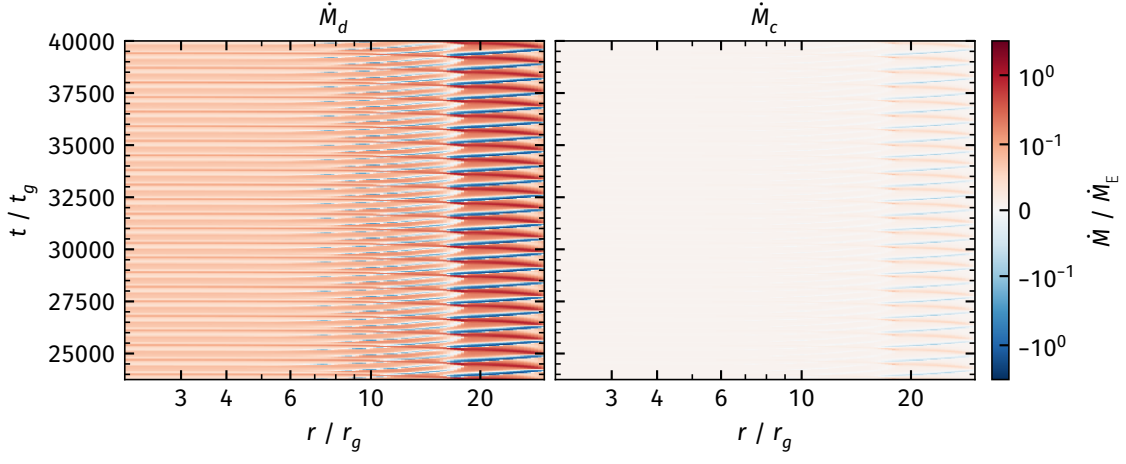
Steady State – Radial Velocity



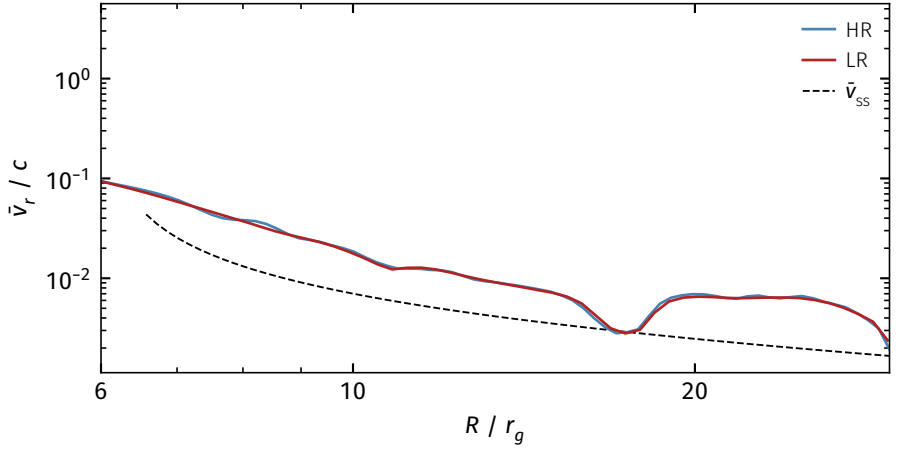
Steady State – Rotation Velocity



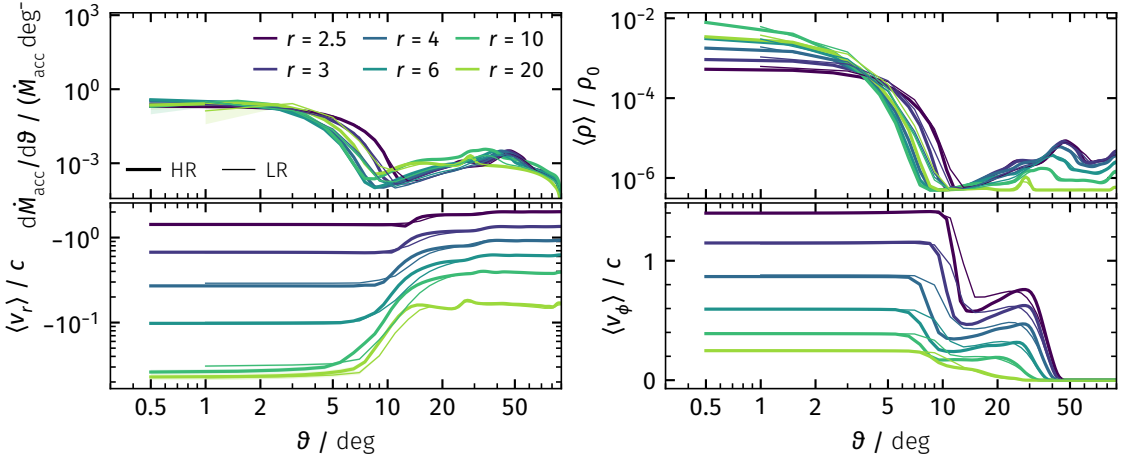
Steady State – Accretion Rate Map



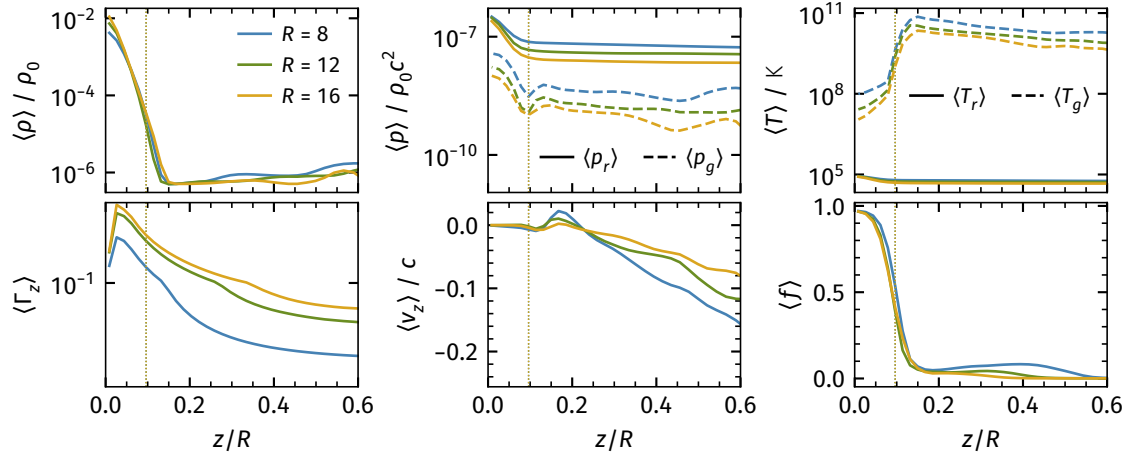
Steady State – Inflow Speed



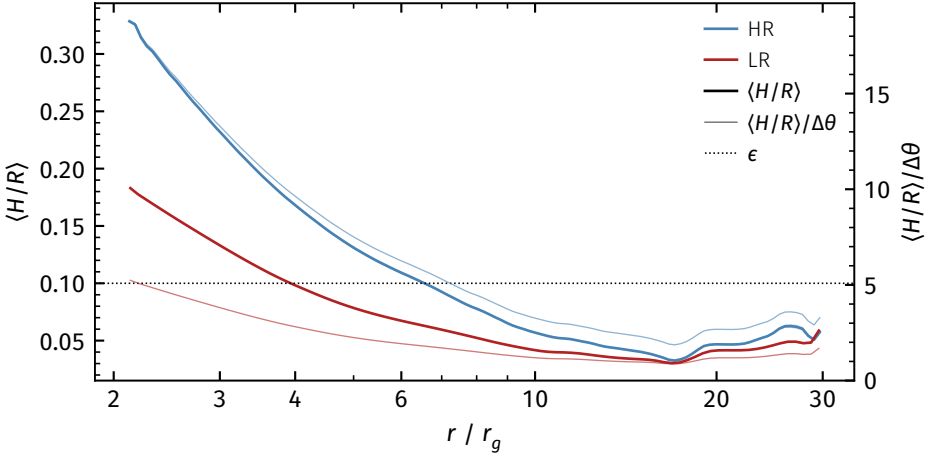
Steady State – Angular Profiles



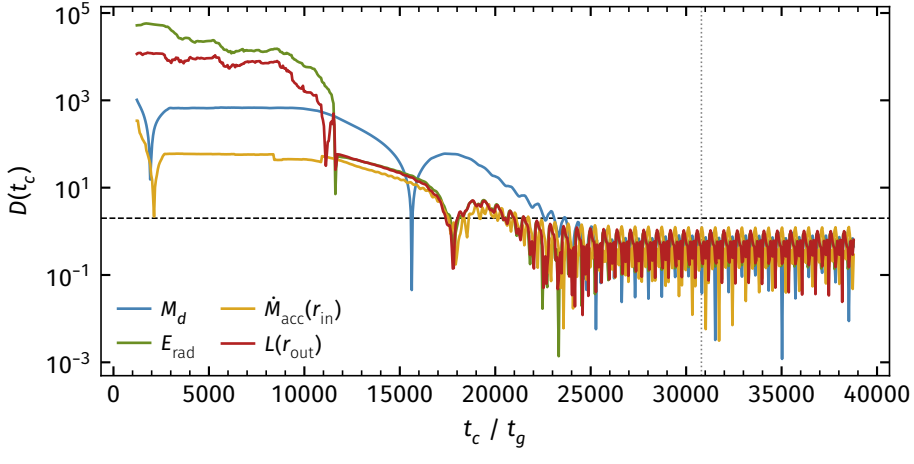
Steady State – Vertical Profiles



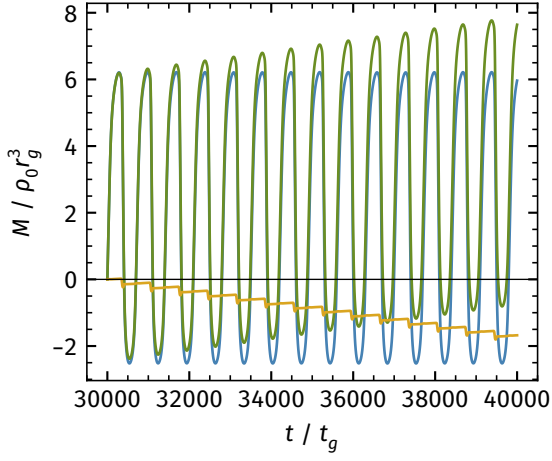
Steady State – Aspect Ratio

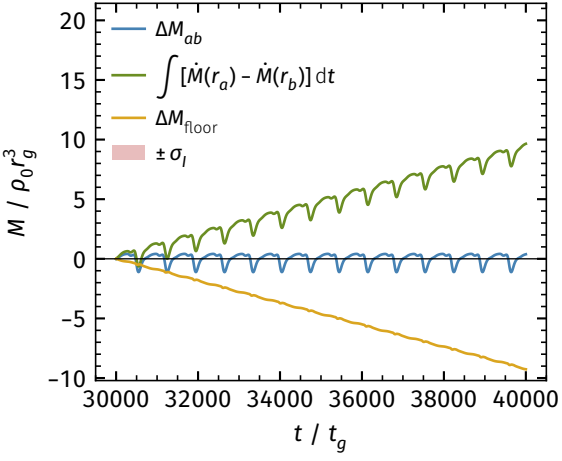


Diagnostics – Relaxation Statistic

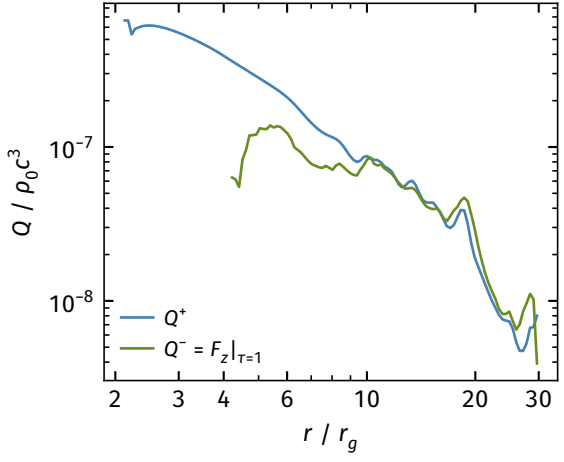


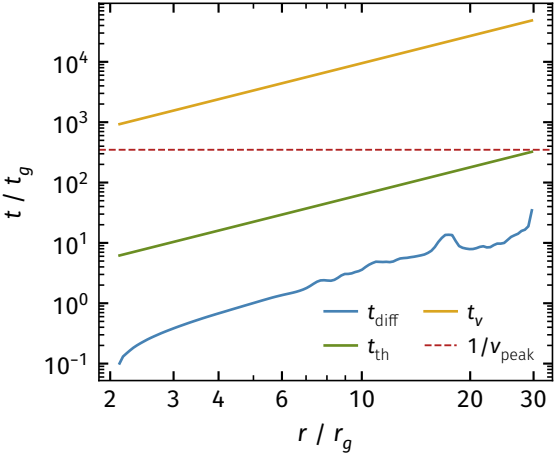
Diagnostics – Mass Budget



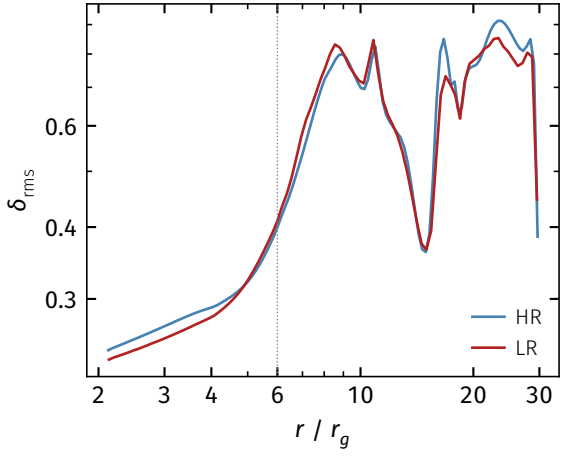


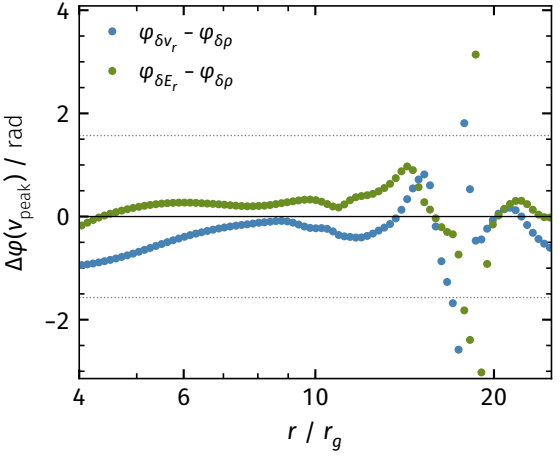
Diagnostics – Energy Balance





Diagnostics – Wave Amplitude & Cross Phase





Diagnostics – Timescales

